

# Supplementary Analyses

## Online influence on a custom social media platform

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## 1 Overview

This document collects the analyses referenced as “Supplementary Analyses” in the manuscript. The original code that generated the analyses in this document is in a quarto document available in the project repository.

## 2 Standardised residuals for the chi-square tests of engagement

The engagement results report several chi-square tests. Here we give the cellwise standardised residuals for each significant omnibus test, which indicate *which* cells drove the effect: a positive residual means the observed count was larger than expected under the null, a negative residual means it was smaller, and  $|z| > 1.96$  corresponds roughly to  $p < .05$ . The tests themselves (and the figures they accompany) are reported in `02_engagement.qmd` of the code and data on the OSF and Github.

### 2.1 Composition of engagement types across conditions

The relative composition of engagement types (favourites, reblogs, replies, and original posts) differed across conditions,  $\chi^2(6) = 279.06$ ,  $p < .001$ . Residuals (rows = condition, columns = engagement type) show that the **Left** condition had more favourites and fewer reblogs than expected, while the **Right** condition showed the opposite pattern (more reblogs, fewer favourites) along with fewer original posts.

|         | Favourite | Reblog | Reply | Post  |
|---------|-----------|--------|-------|-------|
| Left    | 5.54      | -12.27 | 0.17  | 1.13  |
| Control | 1.04      | -3.22  | -0.09 | 1.25  |
| Right   | -6.72     | 15.73  | -0.09 | -2.34 |

### 2.2 Political polarity of engagement across conditions

For each engagement type we tested whether the political polarity of engagement (left, neutral, right) differed across conditions. Residuals (rows = polarity, columns = condition) show, for example, an over-representation of right-wing posts and an under-representation of neutral posts in the **Right** condition, with the strongest effects for reblogs and favourites.

**Post** —  $\chi^2(4) = 9.17$ ,  $p = 0.057$

|         | Left  | Control | Right |
|---------|-------|---------|-------|
| Left    | 0.40  | -0.80   | 0.36  |
| Neutral | 0.54  | 2.05    | -2.54 |
| Right   | -0.98 | -1.17   | 2.15  |

**Reply** —  $\chi^2(4) = 11.47$ ,  $p = 0.022$

|         | Left  | Control | Right |
|---------|-------|---------|-------|
| Left    | 0.65  | -1.31   | 0.56  |
| Neutral | -1.25 | 3.35    | -1.83 |
| Right   | 0.34  | -1.37   | 0.92  |

**Reblog** —  $\chi^2(4) = 38.09, p = 0.000$

|         | Left  | Control | Right |
|---------|-------|---------|-------|
| Left    | 2.24  | -5.09   | 2.49  |
| Neutral | -0.63 | 5.26    | -3.93 |
| Right   | -1.99 | 1.16    | 0.62  |

**Favourite** —  $\chi^2(4) = 118.46, p = 0.000$

|         | Left  | Control | Right |
|---------|-------|---------|-------|
| Left    | 5.99  | -3.20   | -3.14 |
| Neutral | -2.37 | 8.07    | -5.18 |
| Right   | -4.78 | -3.12   | 7.86  |

### 2.3 Friendly versus non-friendly engagement across conditions

The number of friendly versus non-friendly engagements differed across conditions,  $\chi^2(2) = 64.62, p < .001$ . Residuals (rows = condition) show that the **Right** condition had significantly fewer friendly engagements than expected while the **Left** condition had significantly more.

|         | Friendly | non_friendly |
|---------|----------|--------------|
| Left    | 5.71     | -5.71        |
| Control | 2.06     | -2.06        |
| Right   | -7.90    | 7.90         |

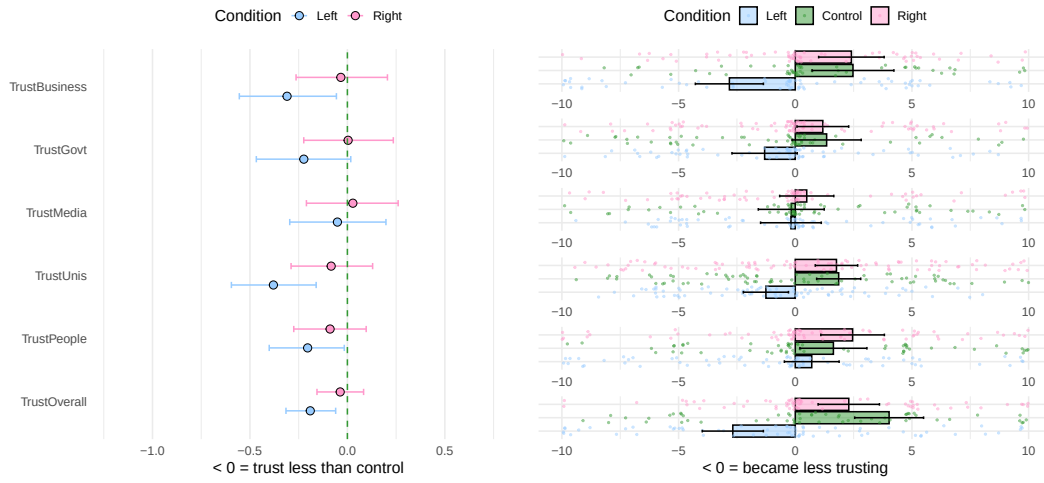
## 3 Linear modelling of each trust, affective polarisation, and relative-belief sub-measure

The main text reports condition effects on the aggregate trust measure (**TrustOverall**), affective polarisation (**AffPol**), and relative-belief (**RelativePolitics**) measures (Figure 3). Here we break those down into their component sub-measures, using the same model structure (**measure After ~ measure Before + Condition**, Bayesian beta regression) as the main analysis.

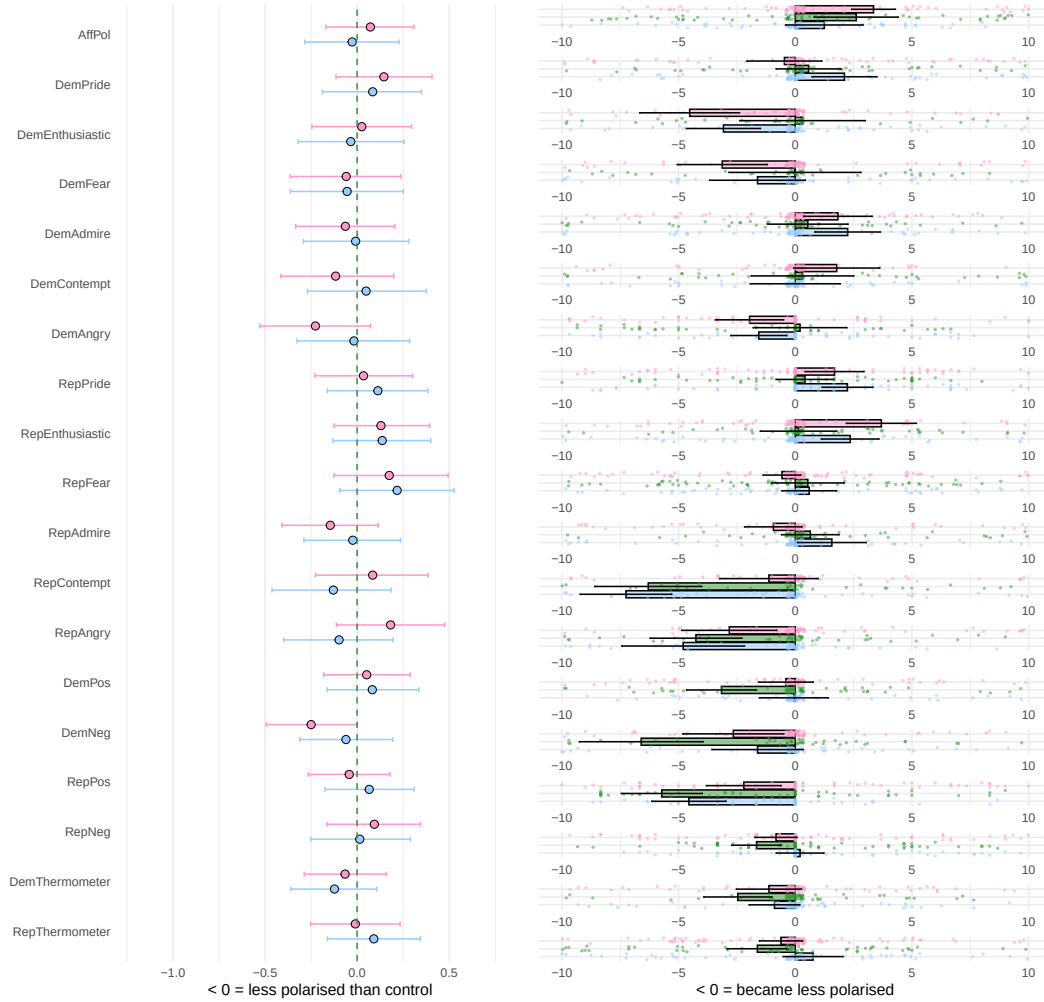
### 3.1 Trust and affective polarisation sub-measures

The trust panel shows that, relative to the **Control**, trust in the **Left** condition decreased credibly for businesses and universities, and that the increase in trust in other people was credibly smallest in the **Left** condition. Trust in the government decreased in the same direction, but the credible interval overlaps zero, and trust in the media showed no credible difference from **Control**.

### Trust Measures

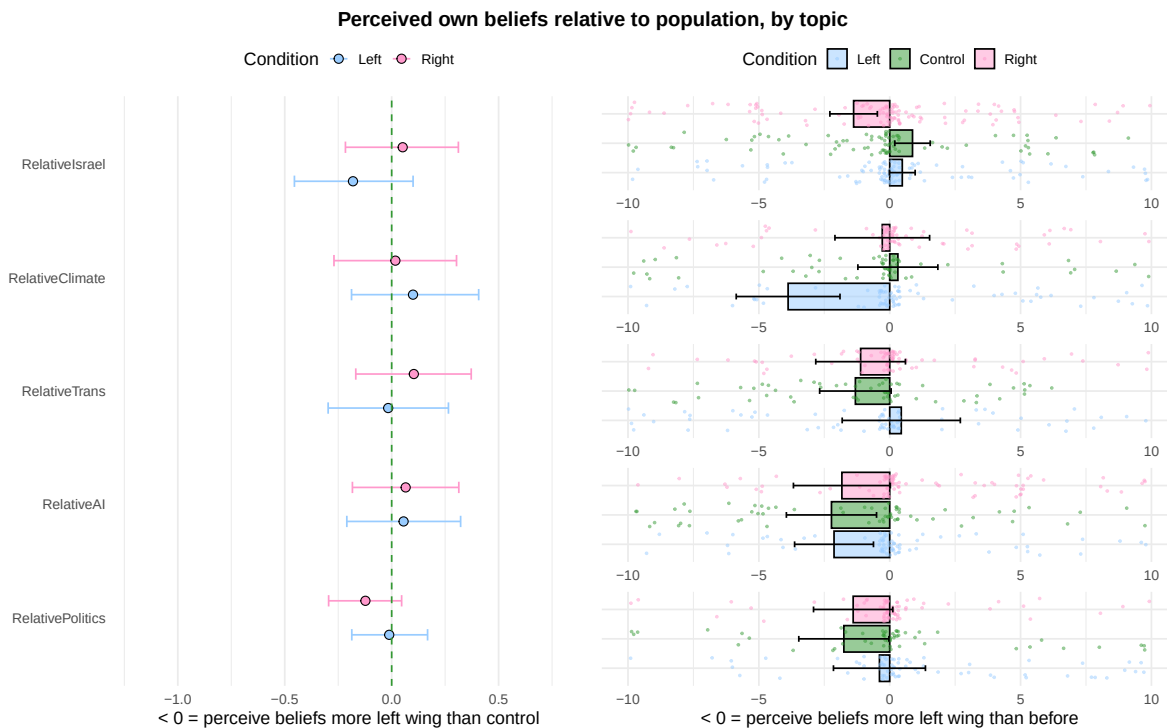


### Affective Polarisation Measures



### 3.2 Relative-belief sub-measures (by topic)

“RelativePolitics” refers to the combined overall relative beliefs collapsed across each subdimension.



## 4 Comparing the full sample to participants who did not have “posts seen” data available

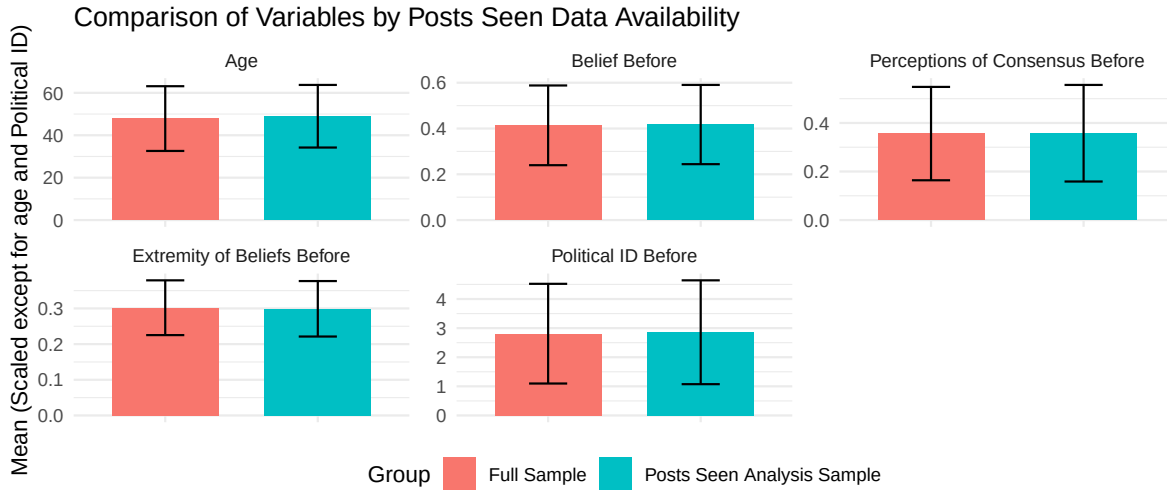
Posts-seen data could not be recorded for every participant (due to technical issues for ~12% in Left, ~23% in Right, ~2% in Control), and the posts-seen analysis was further restricted to participants who saw at least 50 posts. Here we check that the sub-sample with usable posts-seen data did not differ from the full sample on the key baseline measures: initial beliefs (`BeliefBefore`), political identification (`PoliticalIDNum`), extremity of beliefs (`ExtremeBeliefBefore`), perceptions of consensus (`ConsensusPoliticsBefore`), and age. Since we don’t use the control condition in this analysis, we exclude this condition here.

```
[1] "Posts seen data is missing for 53 participants total"
```

```
[1] "Breakdown by condition: "
```

# A tibble: 2 x 5

|   | Condition | N     | Missing | PctMissing | Analysed |
|---|-----------|-------|---------|------------|----------|
|   | <chr>     | <int> | <int>   | <dbl>      | <int>    |
| 1 | Left      | 102   | 21      | 20.6       | 71       |
| 2 | Right     | 116   | 32      | 27.6       | 72       |



Welch t-tests comparing the participants in the posts-seen analysis with the **Left/Right** participants who were excluded (no posts-seen data, or fewer than 50 posts seen):

| Variable                     | M_anal-<br>ysed | M_ex-<br>cluded | n_anal-<br>ysed | n_ex-<br>cluded | t      | p     |
|------------------------------|-----------------|-----------------|-----------------|-----------------|--------|-------|
| Age                          | 48.972          | 45.730          | 142             | 74              | 1.446  | 0.151 |
| BeliefBefore                 | 0.417           | 0.407           | 143             | 75              | 0.394  | 0.694 |
| ConsensusPoliticsBe-<br>fore | 0.358           | 0.353           | 143             | 75              | 0.165  | 0.869 |
| ExtremeBeliefBefore          | 0.299           | 0.308           | 143             | 75              | -0.817 | 0.415 |
| PoliticalIDNum               | 2.858           | 2.716           | 141             | 74              | 0.599  | 0.550 |

The means and standard deviations of the baseline measures are very similar with and without the posts-seen restriction, indicating that the missing posts-seen data did not systematically bias the sub-sample.

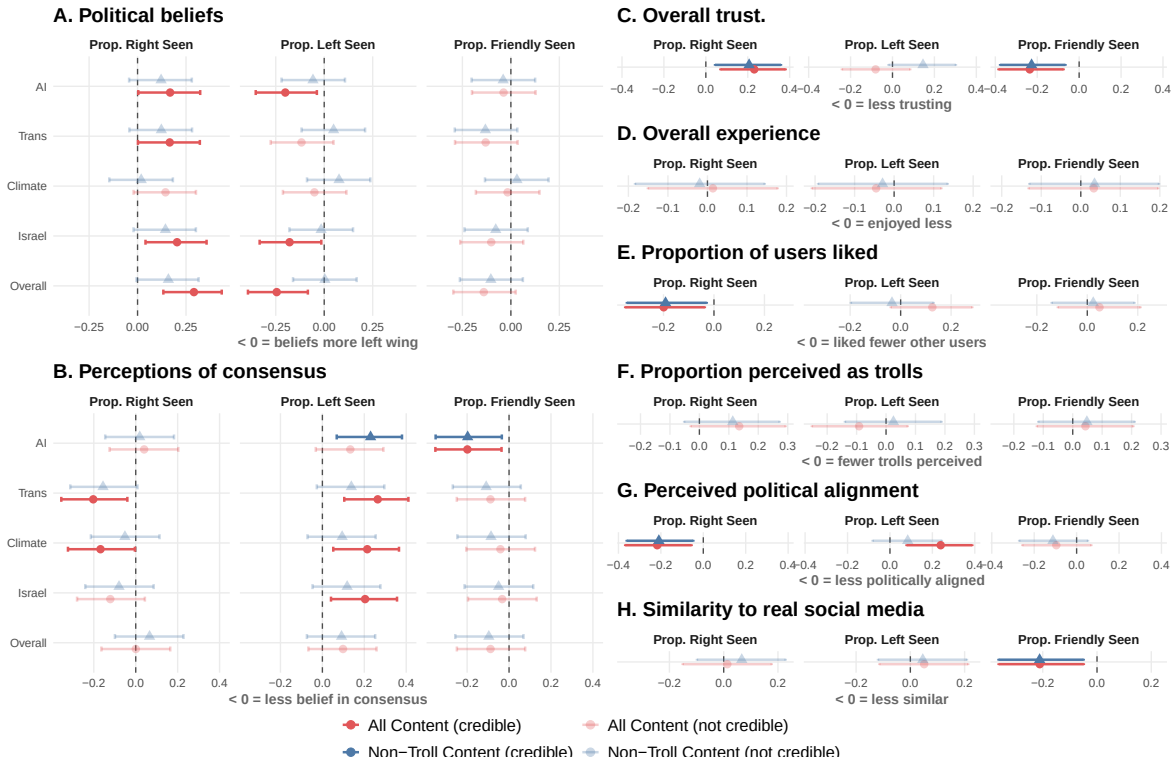
For completeness, the table below reports the mean and variance of the total proportion of political content seen in each condition, for all content, for troll-initiated content, and for non-troll-initiated content.

# A tibble: 2 x 7

|   | Condition | var_troll | mean_troll | var_nontroll | mean_nontroll | var_all | mean_all |
|---|-----------|-----------|------------|--------------|---------------|---------|----------|
|   | <chr>     | <dbl>     | <dbl>      | <dbl>        | <dbl>         | <dbl>   | <dbl>    |
| 1 | Left      | 0.000415  | 0.991      | 0.00405      | 0.464         | 0.00388 | 0.537    |
| 2 | Right     | 0.0143    | 0.942      | 0.00213      | 0.569         | 0.00174 | 0.637    |

## 5 Posts-seen analysis: correlation equivalent of Figure 5

Figure 5 in the main text uses Bayesian beta-regression coefficients to relate the proportion of each post type a participant saw (Right, Left, Friendly) to the change in each outcome measure, comparing all content seen to non-troll-initiated content only. Here we reproduce that comparison using simple correlations rather than the regression models.



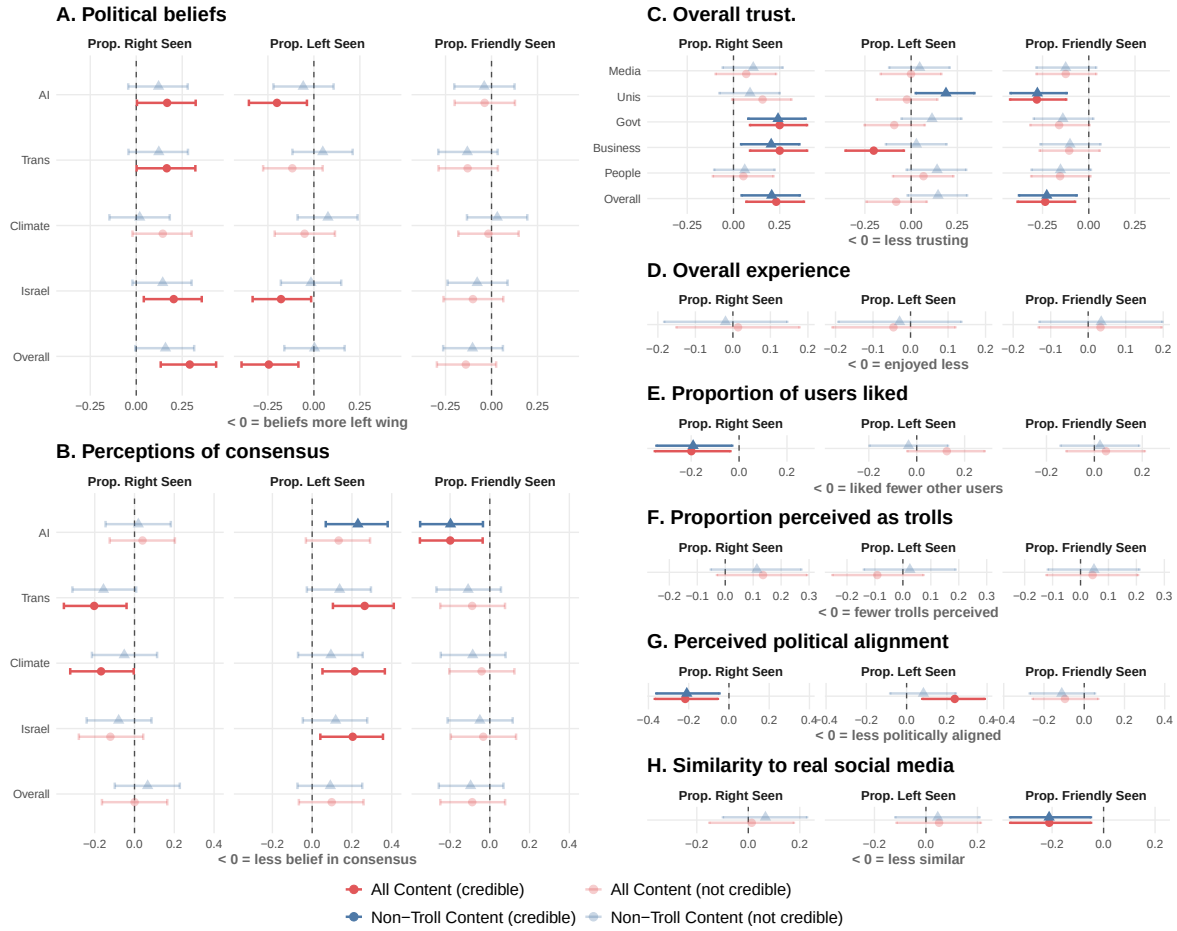
As in the main analysis, the credible correlations between the proportion of each post type seen and the outcome measures closely track the regression results, and credible associations with beliefs and perceptions of consensus disappear once troll-initiated content is removed. Note that there are a few additional credible effects, which may be because these correlations are slightly more powerful (at least for the before/after measures), but they are also less rigorous, so any effects that are not also seen in the model analyses should be interpreted with caution.



## 5.1 Trust sub-measures (correlation and coefficient grids)

The same posts-seen comparison broken down by individual trust sub-measure rather than the aggregate trust measure, shown both as correlations and as the Bayesian regression coefficients from Figure 5.

## 5.2 Correlations



### 5.3 Model Coefficients (same as in manuscript)

